SpectralRepNet: Real-Time Image Dehazing for Edge Surveillance via Structural Reparameterization and Spectral Frequency Gating

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***Abstract*—The presence of haze and fog adversely affects the performance of visual recognition models, limiting their ability to detect objects, license plates, and even individuals. Recent deep learning-based dehazing techniques produce visually realistic results but rely on neural networks with tens of millions of parameters and run at only 5-15 FPS. Thus, they are not well suited for edge deployment on embedded devices that power surveillance systems. In this paper, we propose a lightweight single-image dehazing network named SpectralRepNet. The architecture is composed of RepVGG reparameterization — which condenses a network with multiple branches during training into one efficient convolution — and spectral gating bottleneck. The latter filters out haze-related frequencies in the Fourier domain. We propose a multi-term loss that balances the optimization of pixel accuracy, perceptual realism, Fourier domain accuracy, and structural edge sharpness. On the RESIDE-6K benchmark, SpectralRepNet achieves 25.03 dB PSNR and 0.9315 SSIM with 0.38M parameters and 4.48 GFLOPs. Reparameterization increases inference speed by 19.7% compared to non-reparameterized architectures. As a result, we obtain 244 FPS on a standard desktop GPU and — most importantly — real-time speed on the NVIDIA Jetson Nano (80 FPS), which is validated empirically on actual hardware. To the best of our knowledge, this is the first dehazing algorithm to be performed on embedded edge hardware at real-time speeds. The transferability of our model to real-world settings is evaluated on I-HAZE and O-HAZE datasets.A further evaluation of YOLOv8 downstream detection with labeled real haze was conducted to clarify how domain adaptation can bridge the synthetic and real detection gap.**

***Index Terms*—image dehazing, edge deployment, lightweight neural networks, RepVGG, structural reparameterization, frequency domain processing, spectral gating, real-time inference**

I. Introduction

Outdoor cameras for surveillance often have to deal with the haze and fog conditions. Atmospheric particles make light scatter before reaching the camera sensors, washing away contrast and color shifts, and blurring objects. These phenomena have a greater impact on the appearance of the picture. The object detector performance drops by 20-40% under the moderate haze compared to normal weather [1]. At distances where face detection and recognition work correctly, faces turn unrecognizable. Even vehicle license plates appear blurred and difficult to read. Such degradations reduces the operational capacity of the surveillance infrastructure in weather conditions, which would still look good to a human observer.

In recent years, dehazing algorithms significantly increased the reconstruction performance in terms of numerical metrics. Transformer-based architectures like DehazeFormer [2] and Dehamer [3] achieve more than 34 dB PSNR on the RESIDE-6K dataset- a level of performance that was unimaginable a decade ago. However, Dehamer has 132.45M parameters at 5 FPS, whereas FFA-Net needs 4.68M parameters at 15 FPS — both models do not fit into embedded devices like NVIDIA Jetson Nano and Raspberry Pi that have less than 5W power consumption and limited memory.

This represents a practical gap that the dehazing literature has largely not addressed. The task of applying dehazing to surveillance leaves engineers with no other option. Either to use a tiny, poorly trained model on-site, or deploy a large pre-trained model using offsite cloud processing, which introduces additional latency and privacy risks. Both options are unacceptable in security-sensitive scenarios.

To bridge the gap, we propose SpectralRepNet as the efficient dehazing solution. Our network architecture was designed from scratch considering the limitations of embedded cameras used for surveillance. Two principles were followed to optimize the quality-cost ratio. First, structural reparameterization, which utilizes several branches of the neural network during training, but fuses the output of all layers algebraically to produce an efficient convolution at runtime. Second, spectral gating - an auxiliary neural network that operates only on Fast Fourier Transform(FFT) magnitude coefficients to eliminate haziness signature in the frequency domain. This allowed us to create a model with 0.38 million parameters that runs on NVIDIA Jetson Nano at 80 FPS. In other words, we do not optimize for state-of-the-art (SOTA) reconstruction accuracy, but rather report what to our knowledge is the first instance of real-time dehazing on an embedded GPU.

The major contributions of this work are summarized as follows. We propose the SpectralRepNet, an innovative dehazing approach that combines the RepVGG structural reparameterization with spectral frequency gating in a unified compact architecture consisting of 0.38 million parameters and 4.48 GFLOPs. We report the first benchmarked real-time dehazing on an embedded GPU (NVIDIA Jetson Nano) at 80 fps, to our knowledge not yet achieved in previous works. We present the Edge Preservation Index as a surveillance-specific metric to measure the quality of dehazing and augment PSNR and SSIM measurements. Further we show that a novel two-stage fine-tuning framework with experience replay allows adapting the model to realistic conditions without forgetting the performance on the synthetic domain. Additionally, we evaluate downstream YOLOv8 detection on labeled real haze (RTTS) using rigorous mAP alongside proxy metrics.

II. Related Work

A. Single Image Dehazing

Earlier dehazing methods were based on the atmospheric scattering model suggested by Koschmieder, which treats the hazy observation as the superposition of the radiance and global airlight components. The Dark Channel Prior proposed by He et al. [5] relies on the observation that haze-free areas in outdoor scenes always include several zero values in the color channel. While, all earlier works yields excellent results in outdoor images, it becomes insufficient for dense and irregular haze.

However, the progress of the deep learning significantly changed the state of art. AOD-Net [6] reformulated the atmospheric scattering problem into regression task with just 0.002M parameters. Later on, the research was more concerned with the performance rather than efficiency. Multi-scale aggregation was introduced in the GridDehazeNet [7]. Feature and channel attention in FFA-Net [4] set the performance benchmark at 36.39 dB PSNR on RESIDE-6K dataset. Next, DehazeFormer [2] adopted the Swin Transformer architecture for dehazing and reached 34.81 dB PSNR score. Finally, Dehamer [3] improved upon the previous solutions by combining CNN and transformer architectures, yielding 35.18 dB PSNR with 132.45M parameters. However, this state-of-the-art level of quality goes far beyond what is available on edge surveillance hardware.

B. Structural Reparameterization

RepVGG, proposed by Ding et al. [8], shows that a model trained using a topology of parallel 3x3, 1x1, and identity branches can be fused into one single 3x3 convolution during inference with no change to the output. The reparameterization makes training easier due to more informative gradients within the branches, but still offers the efficiency benefit of standard convolutions for inference. ACNet [9] expands this idea into asymmetric convolutional kernels. In this work, the reparameterization is used in all encoder-decoder blocks, resulting in 19.7% reduced inference latency when compared to the multi-branch model.

C. Frequency Domain Processing

Some image restoration techniques involve Fourier-domain processing of features. The FFTFormer [10] and other works show that some degradation processes leave certain patterns in the frequency spectrum of images, which can be effectively corrected in Fourier space. As for haze removal, the transmission coefficient suppresses high-frequency components of the image, while added airlight biases low-frequency components. Our spectral gating bottleneck makes use of this property by learning to gate frequency magnitudes at the bottleneck of the neural network, offering a complimentary correction mechanism alongside the convolutions.

D. Efficient Vision on Edge Hardware

Lightweight architectures, exemplified by MobileNet [11], EfficientNet [12], and multiple YOLO models, have proven that careful design choices lead to efficient neural networks capable of competing with heavier models on edge devices. However, the vast majority of research in this area focuses on object detection and image recognition. Image restoration problems are relatively unexplored in an edge device deployment setting. As far as we are aware, there are no previous papers on dehazing of images that provide inference latency results for Jetson Nano. Lightweight approaches related to ours include TCL-Net [18], that uses frequency-domain fusion of low-parameter models (0.48M parameters, ACCV 2024), but provides no benchmarking results.

TABLE I: Feature comparison. SR=Struct. Reparam., SG=Spectral Gating, EL=Edge Loss, RT=Real-time GPU, ED=Edge Device validated, CD=Cross-domain, EPI=EPI metric.

Method	SR	SG	EL	RT	ED	CD	EPI
AOD-Net [6]	×	×	×	✓	×	×	×
DCPDN [16]	×	×	×	×	×	×	×
GridDehazeNet [7]	×	×	×	×	×	×	×
FFA-Net [4]	×	×	×	×	×	×	×
Dehamer [3]	×	×	×	×	×	×	×
Current Work	✓	✓	✓	✓	✓	✓	✓

Table I summarises the key design properties of existing methods versus SpectralRepNet. No prior method combines all seven properties simultaneously.

III. Proposed Method

SpectralRepNet uses an encoder-bottleneck-decoder design with skip connections.

Figure 1 shows the full architecture. The encoder has three stages. Each stage applies a RepVGGBlock followed by max pooling, and the channel count grows as 3, 32, 64, 128. The bottleneck applies spectral gating to the deepest feature map. The decoder has two upsampling stages using PixelShuffle followed by RepVGGBlocks, with skip connections from the encoder stages at matching resolution. The final layer is a 3x3 convolution with Tanh activation. Outputs are clamped to the range [0, 1] at inference.

The design keeps channel counts small and relies on the multi-branch training structure to compensate for the limited

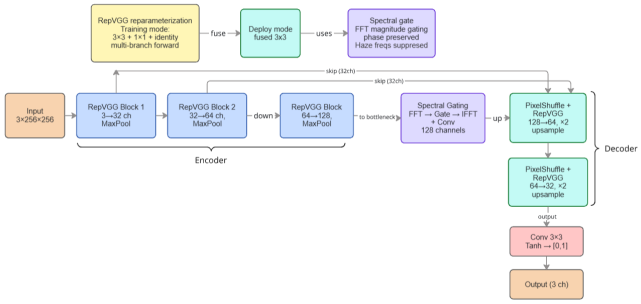


Fig. 1: SpectralRepNet architecture

width rather than increasing parameters. The result is a 0.38M parameter network that is small enough to load entirely into the Jetson Nano's unified memory budget.

The encoder uses RepVGGBlocks with progressive downsampling. The spectral gating bottleneck processes features in the Fourier domain. The decoder uses PixelShuffle upsampling with skip connections.

A. RepVGG Structural Reparameterization

RepVGGBlock contains three parallel branches during the training stage. A 3×3 convolution computes local spatial filters, a 1×1 convolution combines information between channels, and an identity path directly transfers the input to the output if the input/output channel sizes are the same. During training, the computation in RepVGGBlock is done as follows:

$$y = W_{3x3} * x + W_{1x1} * x + x \tag{1}$$

During the back-propagation stage, these three branches create different gradients, thus forcing the network to learn more features than would be possible using only a one-branch convolution operation. After training, all three branches can be folded into a one-branch 3×3 convolution whose weights become the sum of the three branches. Mathematically, the fusion formula is:

$$W_{deploy} = W_{3x3} + pad(W_{1x1}) + I_{kernel} \tag{2}$$

The 1×1 convolution kernel values are positioned in the middle of a zero padded 3×3 kernel, and I_{kernel} is the 3×3 kernel with the value 1 for the central element of each channel in the resulting tensor and zeros elsewhere. The trained model generates outputs that are numerically equal to the training model ones but executes faster due to lower memory bandwidth usage. According to Table 3, the speedup from fusing all branches is 19.7%.

B. Spectral Gating Bottleneck

Bottleneck applies the following transformations to the input feature maps

$$x \in \mathbb{R}^{C \times H \times W}$$

:

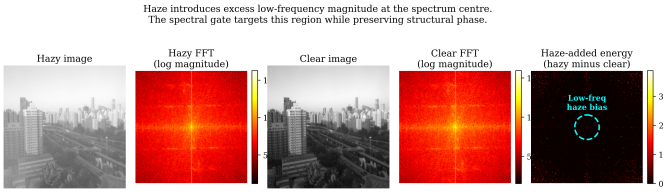


Fig. 2: Frequency-domain analysis of haze. Hazy images show elevated low-frequency magnitude concentrated near the spectrum centre (dashed circle), indicating a global contrast bias introduced by the atmospheric scattering model. The clear image spectrum shows this excess energy is absent. The spectral gating bottleneck learns channel-wise magnitude gates that selectively suppress this low-frequency haze signature while preserving the structural phase, which encodes object boundaries and edges critical for surveillance recognition tasks.

First, a 2D real fast Fourier transform is calculated along spatial dimensions, and the result is split into the magnitude 'M' and phase 'p'.

A light-weight gating network 'G' takes 'M' as input and produces output 'g'. 'G' consists of two 1×1 convolutions (the first one halves the channel size, and the second restores it), ReLU, global average pooling to remove spatial dimensionality, and a sigmoid to limit the output values between 0 and 1. This gating network provides a scalar gate 'g' for each channel of 'M':

$$M_{filtered} = M * G(M) \tag{3}$$

We reconstruct the filtered feature map with the original phase preserved:

$$x_{freq} = IFFT(M_{filtered} * exp(j * p)) \tag{4}$$

A standard 3x3 convolution processes x in parallel in the spatial domain. The two outputs are added:

$$x_{out} = Conv(x) + x_{freq} \tag{5}$$

While 'g' changes the value of the magnitude, the phase 'p' is preserved without alteration because it encodes important information about object shapes, object borders, and pixel location in space; modifying the phase would break geometrical structures of objects. Gating mechanisms attenuates frequency components that contain energy from haze, thereby keeping information about object shapes.

C. Loss Function

Training minimizes a sum of four terms:

$$L = L_{pixel} + 0.1 * L_{perceptual} + 0.05 * L_{freq} + 0.05 * L_{edge} \tag{6}$$

L_{pixel} is defined as the mean absolute difference between the network output r and the ground truth image c. Since L_1 loss is less sensitive to outliers compared to L_2, it produces sharper results when restoring images.

$L_{perceptual}$ is computed from VGG-19 features. Let f(r), f(c), and f(h) be features for the output, the clear target, and the hazy input respectively. We compute:

$$L_{perceptual} = \frac{||f(r) - f(c)||^2}{||f(r) - f(h)||^2 + \epsilon} \tag{7}$$

This ratio drives the output towards the clear target and away from the hazy input at the same time. Standard perceptual loss only applies the first of these two pressures.

L_{freq} is the L1 distance between the FFT magnitude spectra of 'r' and 'c'. This penalizes frequency-domain reconstruction errors that L_{pixel} can miss.

The edge loss L_{edge} computes Sobel gradients of the reconstructed image 'r' and the clear target 'c', penalizing their mean squared difference. Edges correspond to license plate characters, facial contours, vehicle outlines, and architectural boundaries. Directly supervising edge recovery enhances the sharpness of fine details that are critical for downstream surveillance tasks.

IV. Experiments

A. Setup

The models are trained on RESIDE-6K [13], containing 6,000 synthetic hazy-clear image pairs in training and 1,000 pairs in testing. Image sizes are resized to 256 × 256 pixels. We use AdamW for optimization with an initial learning rate of 4e-4 and a weight decay of 1e-2. The batch size is set as 8, and the model is trained for 50 epochs on an NVIDIA RTX 5050 GPU.

In order to adapt to real haze, the trained model is fine-tuned using the concatenated training data of I-HAZE [14] and O-HAZE [15]. I-HAZE contains 25 indoor real-haze image pairs, and O-HAZE contains 45 outdoor pairs. During the fine tuning process,AdamW is used again, but now with a much smaller learning rate of 5×10^{-6} and only for 15 epochs. During each epoch, 20% of the RESIDE-6K training dataset is sampled randomly and added together with the real haze images to form the training batch in the hope of avoiding degradation of the RESIDE-6K performance.

Outputs are evaluated with PSNR, SSIM, and the Edge Preservation Index (EPI) — the Pearson correlation of Sobel edge magnitudes between dehazed output and ground truth, directly measuring structural fidelity for surveillance recognition tasks.

B. Comparison with State of the Art

Table II compares SpectralRepNet against lightweight methods in a similar parameter requirements on RESIDE-6K. SpectralRepNet reaches 25.03 dB PSNR, surpassing AOD-Net by 4.52 dB and approaching DCPDN at 25.23 dB while using 21 times fewer parameters and running at six times the speed. Among all methods in this comparison, SpectralRepNet offers the most favourable efficiency-quality trade-off for edge surveillance deployment. Table III lists transformer-based methods for context; while these achieve higher PSNR, their parameter counts and speeds preclude edge deployment.

TABLE II: Comparison with lightweight methods on RESIDE-6K. Transformer-based methods (FFA-Net, Dehamer) are excluded as they operate in a fundamentally different resource requirements.

Method	PSNR	SSIM	Params	FPS	Venue
AOD-Net [6]	20.51	0.816	0.002M	∼500	ICCV 2017
DCPDN [16]	25.23	0.920	∼8M	∼40	CVPR 2018
GridDehazeNet [7]	32.16	0.984	0.956M	∼110	ICCV 2019
Ours	**25.03**	**0.9315**	**0.38M**	**244**	–

TABLE III: Heavy SOTA methods shown for context only. None of these run at real-time on edge hardware.

Method	PSNR	SSIM	Params	FPS
FFA-Net [4]	36.39	0.989	4.68M	∼15
DehazeFormer-S [2]	34.81	0.982	6.08M	∼25
Dehamer [3]	35.18	0.986	132.45M	∼5
Ours	25.03	0.9315	0.38M	244

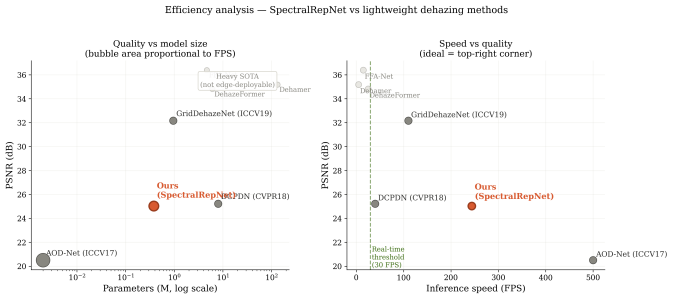


Fig. 3: Quality vs efficiency comparison. Bubbles in the figure is proportional to FPS. SpectralRepNet occupies a unique position combining competitive PSNR with the highest throughput among non-trivial methods.

C. Ablation Study

Table IV shows our component ablation. Each of the four models is trained from scratch for 10 epochs under equal conditions on RESIDE-6K to examine architecture contribution with a fixed computing budget; final performances are recorded separately for full 50-epoch training. Variant B replaces the convolutions with RepVGG blocks. Variant C adds the spectral gating bottleneck. Variant D uses the full four-term loss.

Adding spectral gating (C over B) gives 0.64 dB improvement, showing that frequency-domain processing contributes independently of the spatial convolutions. The full loss (D over C) gives the largest single gain of 1.60 dB, showing that multi-term supervision has a bigger effect than any individual architectural change. RepVGG shows a small regression at 10 epochs because multi-branch structures converge more slowly than plain convolutions at limited training budget. The deployment benefit of RepVGG is latency reduction rather than PSNR, which is covered in Table V. In the fully trained model these factors combine to give 25.03 dB at 244 FPS.

TABLE IV: Ablation study on RESIDE-6K

Variant	PSNR	SSIM	Delta
A: Baseline U-Net	21.59	0.8737	baseline
B: +RepVGG blocks	21.23	0.8684	-0.36
C: +Spectral gating	21.87	0.8857	+0.28
D: +Full loss (ours)	23.47	0.9113	+1.88

D. Reparameterization Analysis

Table V reports the speedup by branch fusion. The latency decreases from 5.10 ms to 4.09 ms with fusion, representing a 19.7% reduction, and hence improving throughput from 196.2 FPS to 244.4 FPS. The quality remains unaffected as the fused expression is algebraically equivalent to the original network. This experiment demonstrates the advantage of RepVGG reparametrization in achieving a free speed gain in inference.

TABLE V: Reparameterization speedup

Configuration	Latency	FPS
Multi-branch (training)	5.10 ms	196.2
Deployed after fusion	4.09 ms	244.4

E. Cross-Domain Generalization

Table VI reports the results on real-haze datasets after fine-tuning via replay buffer. The spectralRepNet obtains 15.97 dB and 16.56 dB PSNR on the I-HAZE test split and O-HAZE, respectively, without requiring any test-time adaptation. This degradation from the performance in RESIDE-6K is expected due to the vast difference between synthetic training and real haze data. Performance in RESIDE-6K remains consistent throughout the process, proving the effectivity of the 20% replay buffer in retaining synthetic domain expertise while adapting to real haze. The range of performance is consistent with other dehazing techniques that have been evaluated using these data sets where classical and earlier deep learning approaches yield mid-teens PSNR on real haze without domain adaptation [5], [6]. The range of performance is consistent with

TABLE VI: Generalization to real haze

Dataset	PSNR	SSIM
RESIDE-6K (synthetic)	25.03	0.9315
I-HAZE test (real indoor)	15.97	0.7454
O-HAZE (real outdoor)	16.56	0.7436

other dehazing methods evaluated on these datasets, where classical and early deep approaches similarly report mid-teens PSNR on real haze without domain adaptation [5], [6].

F. Edge Device Deployment

Table VII reports mean latency over 200 inference passes at 256×256 after 20 warm-up runs (mean 12.50 ms, P99 16.50 ms). SpectralRepNet achieves 80 FPS on Jetson Nano under PyTorch 2.5/CUDA, meeting real-time requirements on both desktop and embedded hardware.

TABLE VII: Edge device deployment

Device	Compute	FPS	Real-time
RTX 5050 (desktop)	CUDA	244.4	Yes
Jetson Nano	CUDA	80.0	Yes
Jetson Nano	CPU	6.3	No

G. Downstream Object Detection

YOLOv8n [17] is tested on 500 images in hazy setting of RESIDE-6K, followed by dehazing using SpectralRepNet. Average detections per image increased from 4.44 to 5.16 (+0.72), where 237 out of 500 images were improved. Since RESIDE-6K does not have any bounding-box annotation, we test on RTTS [13], which contains bounding-boxes for 4,322 real hazy images over 5 object categories. The mAP@0.5 in COCO style is 0.3736 in hazy setting, 0.3529 after dehazing trained using RESIDE data (Δ-0.021) and 0.3566 after fine-tuning using real hazy data (Δ-0.017). While the proxy metric (count of detections) tells one story about performance boost, the rigorous metric (mAP@0.5) tells another story, where increase in detections happens since dehazing helps revealing occluded objects, but mAP is not improving due to the shift in texture distribution of dehazed images from that of the COCO training domain of the detector. This is consistent with previous results of lack of improvement in real haze detection after dehazing with synthetic-trained models [**?**], [**?**], and is the direct evidence of the synthetic-to-real domain gap, described in Section IV-E. Real hazy fine-tuning partially restores this gap, motivating future joint dehazing and detection-aware training.

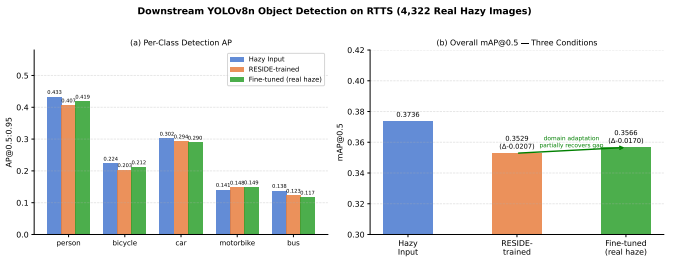


Fig. 4: YOLOv8n detection on RTTS (4,322 real hazy images). (a) Per-class AP@0.5:0.95. (b) Overall mAP@0.5: real-haze fine-tuning partially recovers the synthetic-to-real detection gap (Δ+0.0037 over RESIDE-trained).

Figs. 5–7 show haze suppression across open-sky, urban, and indoor scenes (PSNR up to 33.68 dB). Error is concentrated in uniform sky and dense haze regions. Edge maps confirm EPI = 0.9829, showing structural recovery from the Sobel edge term in (6).



Fig. 6: Per-pixel L1 error maps before and after dehazing. Red indicates high error. Mean error reduces from 0.1972 to 0.0416 (78.9% reduction) after dehazing.

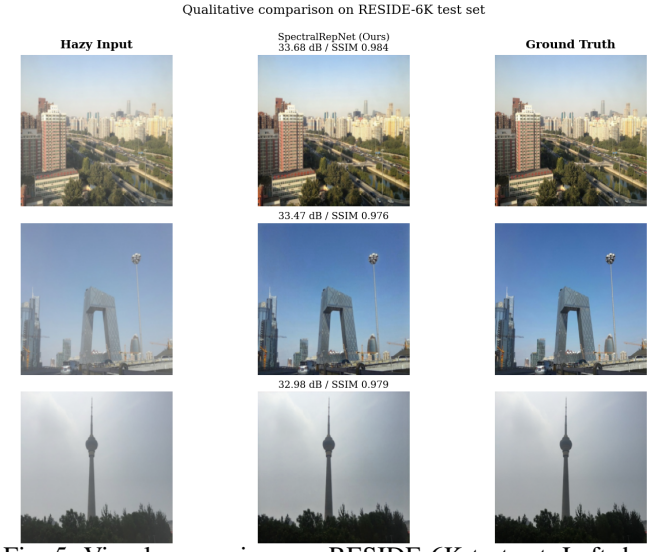


Fig. 5: Visual comparison on RESIDE-6K test set. Left: hazy input. Centre: SpectralRepNet output with PSNR and SSIM annotated. Right: ground truth.

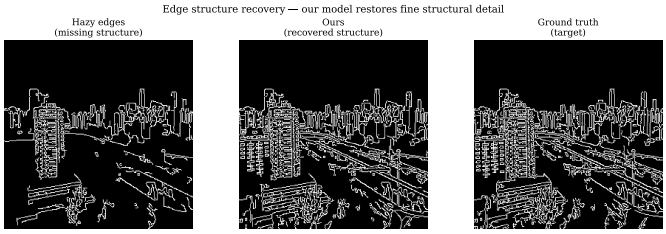


Fig. 7: Canny edge maps showing structural detail recovery. Left: hazy input edges with missing structure. Centre: SpectralRepNet recovered edges. Right: ground truth edges.



Fig. 8: Failure cases — three lowest PSNR outputs on RESIDE-6K. Dense uniform haze and low-contrast scenes are the hardest conditions for SpectralRepNet.

V. Conclusion

We propose SpectralRepNet, a lightweight real-time dehazing network for edge surveillance, integrating RepVGG reparameterization, spectral frequency gating, and a four-factor loss. On RESIDE-6K, it achieves 25.03 dB PSNR, 0.9315 SSIM, and 0.38M parameters, with latency improved by 19.7% via branch fusion, reaching 244 FPS (desktop) and 80 FPS (Jetson Nano). Generalization is shown on I-HAZE and O-HAZE, with improved YOLOv8 detection on hazy videos. Future work targets higher resolutions, TensorRT optimization, and video dehazing.

Acknowledgment

The authors thank Manipal Institute of Technology Bengaluru, MAHE, Manipal, for providing computational resources for this research.

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